



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

Name: Aman S. Choudhary

PRN: 25070122033

Div: CSE A1

Batch:2025-29

ASSIGNMENT No : 9

TITLE :Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Outcome

The program demonstrated the use of the final keyword with a variable to prevent reassignment after initialization. Attempting to modify a final variable resulted in a compile-time restriction, confirming its immutability behavior.

Conclusion

The final keyword is useful for defining constants and protecting critical data such as account numbers or IDs from accidental modification, thereby improving the reliability and security of a Java program.

Exercise

1. Create a **Bank Account** program where account number is final and cannot be changed.

```
1  class BankAccount {
2      final int accountNumber;
3      double balance;
4      BankAccount(int accountNumber, double balance) {
5          this.accountNumber = accountNumber;
6          this.balance = balance;
7      }
8      void display() {
9          System.out.println("Account No: " + accountNumber + ", Balance: " + balance);
10     }
11 }
12 public class BankAccountDemo {
13     Run | Debug
14     public static void main(String[] args) {
15         BankAccount acc = new BankAccount(accountNumber: 101, balance: 25000.0);
16         acc.display();
17     }
18 }
```

Account No: 101, Balance: 25000.0

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```

1  class Book {
2      final String isbn;
3      String title, author;
4      double price;
5      Book(String isbn, String title, String author, double price) {
6          this.isbn = isbn;
7          this.title = title;
8          this.author = author;
9          this.price = price;
10     }
11     void display() {
12         System.out.println("ISBN: " + isbn);
13         System.out.println("Title: " + title);
14         System.out.println("Author: " + author);
15         System.out.println("Price: " + price);
16     }
17 }
18 public class LibraryDemo {
19     Run | Debug
20     public static void main(String[] args) {
21         Book b = new Book(isbn: "978-3-16-148410-0", title: "Java Basics", author: "J. Smith", price: 450.0);
22         b.display();
23     }

```

```

ISBN: 978-3-16-148410-0
Title: Java Basics
Author: J. Smith
Price: 450.0

```